ZEERBOX · PRODUCT REQUIREMENTS AND SPECIFICATIONS

ZeerBox product requirements and specifications

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-24	Amish Chadha	Kit 1.9.1 upgrade: ZBX-REQ-001 becomes ZBX-PRS-001; R1 to R12 migrated to PR-001 to PR-012 with the same targets, each traced to a user requirement in ZBX-URS-001, with tolerance, units and verification TRL; PR-009 names the 5 A load fuse (ZBX-DEC-001 D1); PR-012 at the target after it
0.5	2026-10-02	Amish Chadha	Status from ZBX-CAL-001 v0.3 after the design for construction (ZBX-DDR-003); R12 reported against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R3 relaxed to 80 %; status from ZBX-CAL-001 v0.2 (150 mm pad, dry rice husk fill)
0.3	2026-09-25	Amish Chadha	Redefine R12 as the cooling equipment kit (ZBX-DDR-001 item 1); R7 on the decided 150 W panel; TRL 3 status from ZBX-CAL-001

0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, design point and status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold (as ZBX-REQ-001)

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What the design must do, each traced to a user need in ZBX-URS-001 and checked by calculation in ZBX-CAL-001 v0.4 on the constructable design (option A, ZBX-OPT-001). Seven are met on paper (three of them by design or logic review only), the kit cost (PR-012) is at its value-engineering target, **PR-003 is at risk**, none is not met, and PR-006, PR-010 and PR-011 can only be verified at TRL 4. Targets are proposals and are not yet validated with users; they will be revised after co-design sessions with the partner (ZBX-PRB-001). The "Was" column gives the requirement's number in ZBX-REQ-001, which the calculation note still uses. Method: STANDARDS section 22.

The **design point** used throughout is outside air at 35 °C and 30 % RH (wet bulb 21.5 °C), a full store of 480 kg of mixed vegetables, and 100 kg of fresh produce loaded per day at 32 °C.

Table 1. Product requirements and specifications. Values are central estimates from ZBX-CAL-001 v0.4, with the favorable to unfavorable range where it matters.

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	W
PR-001	UR-001, UR-002	Physical architecture	Storage capacity and reach	The store shall hold standard 20 kg crates on shelves, every crate reachable by one adult walking in, with no crate lifted above 1.6 m	20 crates (400 kg); top shelf 1.6 m	min; max	crates; m	Analysis of the model (CAL-001 s2), TRL 3; loading trial, TRL 4	Met on paper: 24 crates, 480 kg; top shelf 1.40 m	R1
PR-002	UR-003	Performance	Cooling at the design point	Mean store air shall be below outside air at the design point by at least the target	8	min, unfavorable case	K	Psychrometric and heat balance (CAL-001 s4), TRL 3; logged field data, TRL 4	Met on paper: 10.7 K (9.8 to 11.4 K); store 24.3 °C	R2
PR-003	UR-004	Performance	Humidity for produce	Mean store RH while cooling shall reach the target; leafy and water-sensitive produce goes in covered or lined crates	80	min, unfavorable case	% RH	Calculation (CAL-001 s5), TRL 3; logged field data, TRL 4	At risk: 81 % (77 to 85 %)	R3
PR-004	UR-005	Software	Humid-weather detection and mode lamp	The controller shall detect outside air too humid to cool and switch to night ventilation or hold, showing the mode on a lamp by the door	Switch below 4 K wet-bulb depression	+-1 K sensing error	K	Logic review (CAL-001 s6), TRL 3; bench test with sensors, TRL 4	Met by logic review: 30 °C and 75 % RH gives 3.7 K and switches to hold; sensing error 0.5 K (STM S32)	R4
PR-005	UR-006	Safety	Mold and air-quality guard	The pad pump shall stop if store RH stays above 95 % for 60 min, and the store air shall change at least once an hour whenever produce is inside	1	min	air changes per hour	Logic review (CAL-001 s6), TRL 3; bench test, TRL 4	Met by logic review: hold mode about 8 air changes an hour at about 2	R5

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	W
									W; with no power for more than about 40 h the fans stop (STM S36, ZBX-DEC-001 D3)	
PR-006	UR-007	Performance	Shelf life	Tomato shelf life shall be at least the target multiple of ambient storage in the dry season (for example 6 days to 9 days or more)	1.5	min	x ambient	Literature comparison, TRL 3; paired storage trial with the partner, TRL 4	To verify at TRL 4: ZECC data (about 1.8 times) come from a cooler, more humid chamber	R6
PR-007	UR-008	Electrical	Solar operation	Fans, pump and controller shall run from one PV panel with no grid for a full design day: 10 h of cooling plus 2 h of evening ventilation	372	min, energy available	Wh/day	Energy balance (CAL-001 s7), TRL 3	Met on paper: 525 Wh from the 150 W panel (41 % margin)	R7
PR-008	UR-009	Performance	Water use and sump	Water use per design day, pad and any wall cavity included, shall not exceed the target; the sump shall hold at least one day of pad and bleed water	70	max	L/day	Water balance (CAL-001 s8), TRL 3	Met on paper: about 43 L (no cavity water; 11 L bleed, ZBX-DEC-001 D5); 60 L sump for 43 L	R8
PR-009	UR-010	Safety	Low voltage and safe	All electrical parts shall run at 12 V DC nominal, the battery leads fused at the	12 V DC; fuse at or below the derated	as stated	V; A	Design review (CAL-001 s9), TRL 3	Met by design review: 15 A battery fuse on	R9

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	W
				terminal and the load circuit fused within its cable's capacity; fans guarded both faces; the door shall open from inside without a key	cable capacity				2.5 mm² leads, 5 A load fuse on the 1.5 mm² load circuit, both DC rated (ZBX-DEC-001 D1, D5)	
PR-010	UR-011, UR-012	Physical architecture	Local build and repair	Walls, roof, door and shelves shall be built by a local mason and carpenter; every fan, pump, sensor and pad shall be replaceable with generic parts in the target time	30	max per part	min	Design review, TRL 3; parts survey and swap trials with the partner, TRL 4	To verify at TRL 4	R1
PR-011	UR-013	Materials	Durability	The structure shall last the target life; pad, fans and pump at least 3 years with cleaning, in dust	10	min, structure	years	Material review and structural checks (CAL-001 s10), TRL 3; supplier data and field life, TRL 4	To verify at TRL 4; roof footings 1.8 times the uplift at 30 m/s; beams and purlins sized	R1
PR-012	UR-014	Cost	Affordable kit	The cooling equipment kit (pad, sump and pump, fans, controller, solar panel, power box, wiring and panel frames: BOM items 6 to 11, 14 and 18) shall be priced against the value-engineering target in `project.yaml`; the structure (items 1 to 5, 12,	300	value-engineering target, not a limit	USD	Priced BOM (`bom/bom.csv`), CAL-001 s11, TRL 3	At the target: kit USD 300; structure USD 574, reported separately	R1

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	W
				15, 16 and 17) is built from local materials and costed separately						

Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

PR-012 (R12) was redefined on 2026-09-25 by Amish's decision to go with the TRL 2 recommendation (ZBX-DDR-001 item 1). Its earlier target, the complete store for $300, was not met at about $575. Since 2026-10-01 the USD 300 figure is a hypothetical value-engineering target, not a limit (Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"), so PR-012 is reported as over, under or at the target.

PR-003 (R3) was relaxed from 85 % to 80 % on 2026-09-25 by Amish's decision to go with the TRL 3 recommendation (ZBX-DDR-002, item 11), together with a 150 mm pad and produce-level measures (covered or lined crates). At 85 % it would still not be met: the store averages 81 % and 85 % needs a pad about 185 mm deep or more.

Assumptions

- Direct evaporative pad effectiveness of 75 % for a 100 mm cellulose pad at 0.6 m/s face velocity, scaled for velocity and depth in ZBX-CAL-001; about 88 % for the decided 150 mm pad at 0.52 m/s. Manufacturer data for deeper pads show 90 % or more (CELdek specification); 70 to 80 % is assumed for 100 mm.
- The fans are taken as a straight-line curve from 400 m³/h free air to a 50 to 70 Pa shut-off; pad pressure drop is 15 Pa at 1.0 m/s. Both need supplier data (STEM check S3 and S5).
- The heat load at the design point is 443 W (376 to 559 W), from the heat balance in ZBX-CAL-001.
- Dry rice husk in the wall cavity conducts about 0.06 W/(m·K) and is kept dry and pest-free; measured values of 0.065 to 0.075 W/(m·K) add about 15 W of wall gain (STEM check S23).
- Dry-season solar resource of about 5 peak sun hours, with 30 % combined losses for heat, dust, PWM charging and wiring.
- Tomato shelf life gains are taken from zero energy cool chamber trials, which run cooler and more humid than a forced-air store; PR-006 may be optimistic.
- Recommended produce conditions follow the UC Davis Postharvest Center (tomato 12.5 to 15 °C for mature green, 90 to 95 % RH). ZeerBox cannot reach these temperatures in the design case; PR-002 is set on temperature drop, not absolute temperature.

Requirements not met or at risk at TRL 3

- **PR-003 (humidity), at risk:** the 150 mm pad supplies air at 87 % RH, and the store averages 81 % in the central case but 77 % in the unfavorable case, against 80 %. The air loses about 4 to 5 percentage points of RH per kelvin as it warms across the store. Covered or lined crates hold humidity near the produce; field data will show whether they are enough.

- No requirement is not met. PR-005 holds only while the battery has charge: after about 40 h with no sun the fans stop and the room is sealed (STEM check S36); a passive vent for that case is proposed to Amish (ZBX-DEC-001 D3).